(full-function device) and ZigBee end device. Various network topologies used by ZigBee are shown in Figs 1 through 3.

ZigBee coordinator. The coordinator is the most capable device that maintains the overall network knowledge. It forms the root of the network tree and might bridge to other networks. There is exactly one ZigBee coordinator in each network. It is able to store information about the network, acting as repository for security keys. It is the most sophisticated of the three types and requires the maximum memory and computing power.

ZigBee router. Routers act as intermediate devices, passing data from other devices.

ZigBee end device. The ZigBee end device carries limited functionality, lowering the cost and complexity. It is generally found in network edge devices. The device contains just enough functionality to talk to its parent node (either the coordinator or a router); it cannot relay data from other devices. It requires the least amount of memory and is less expensive to manufacture than the ZigBee coordinator or router.

Traffic types

ZigBee employs either of beacon and non-beacon modes to enable to-and-fro data traffic. The beacon mode is used when the coordinator runs on batteries and thus offers maximum power savings, whereas the non-beacon mode finds favour when the coordinator is mains-powered.

In beacon-enabled networks, special network nodes called ZigBee routers transmit periodic beacons to confirm their presence to other network nodes. Beacon-oriented networks use guaranteed time slots. That means nodes are active only when a beacon is being transmitted. The nodes may sleep between beacons, which lowers the duty cycle and increases the battery life.

In non-beacon-enabled networks, power consumption is asymmetrical: some devices are always active, while others are sleeping. The receivers of ZigBee routers are active continuously, requiring a robust power supply. In non-beacon-enabled networks, nodes talk in the same way as people converse; these briefly check to see that no one is talking before they start.

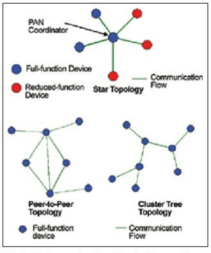


Fig. 1: Various networking topologies used by ZigBee

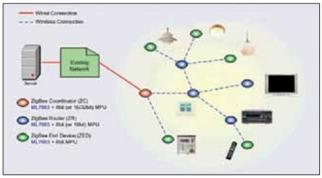


Fig. 2: ZigBee network

Comparison with Bluetooth

Bluetooth is an open wireless technology standard for exchanging data over short distances. It too eliminates cabling between electronics products, such as between computers and printers or between mobile phones and headsets. But Bluetooth is oriented more towards user mobility, whereas ZigBee aims at grand-scale automation and remote control.

Low latency is an important feature of ZigBee. When a ZigBee device is powered off, all the circuitry is switched off, except the clock which runs at 32 kHz. It wakes up and sends packets across the network

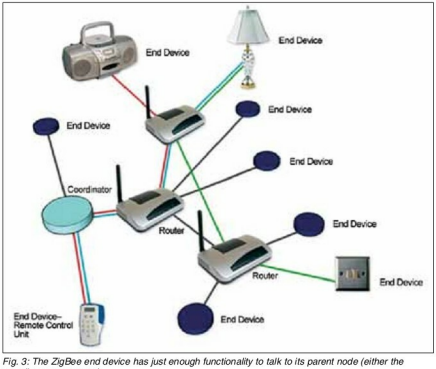


Fig. 3: The ZigBee end device has just enough functionality to talk to its parent node (either the coordinator or a router)